Name: Varsha G Shenoy

Retail Demand Forecasting & Inventory Replenishment Planner
Part A - Problem Framing Document

Business Objective

The objective of this project is to reduce stockouts while controlling inventory holding costs across multiple retail stores. Stockouts lead to lost sales and reduced customer satisfaction, while excessive inventory increases storage and holding costs.

This project aims to improve inventory decisions by combining demand forecasting and data-driven replenishment policies at the store–SKU level. By accurately forecasting demand and optimizing reorder points and safety stock, the system helps ensure that products are available when customers need them while avoiding unnecessary overstock.

North Star Metric

Service Level (Fill Rate)

The North Star Metric for this project is Service Level (Fill Rate).
Fill rate measures the proportion of customer demand that is successfully fulfilled from available inventory.
Fill Rate = Units Sold / True Demand

Fill rate is chosen because:
- It directly reflects product availability to customers
- It captures the impact of stockouts on sales
- It aligns closely with the business goal of reducing lost sales
- It provides a clear measure of inventory service performance across stores and SKUs

Improving the fill rate indicates that the replenishment strategy is effectively ensuring sufficient inventory availability.

Supporting KPIs

To evaluate operational performance and monitor the effectiveness of the replenishment strategy, the following supporting KPIs are used -

Forecast Accuracy

Measured using WAPE (Weighted Absolute Percentage Error).

Forecast Accuracy = 1−WAPE

This metric evaluates how closely the demand forecast matches actual demand.

Stockout Rate

The proportion of time when inventory levels reach zero.

Stockout Rate = Stockout Days / Total Days

Lower stockout rates indicate better product availability.

Fill Rate

The percentage of customer demand that is fulfilled from available inventory.

Fill Rate = Units Sold / True Demand

This is also used as the **North Star metric**.

Inventory Turns (Proxy)

Inventory turns estimate how efficiently inventory is utilized.

Inventory Turns = Total Demand / Average Inventory

Higher inventory turns indicate better inventory utilization.

Days of Inventory on Hand (DOH)

Measures how many days current inventory can satisfy expected demand.

DOH = On Hand Units / Average Daily Demand

Lower DOH suggests faster inventory movement.

Overstock Rate

Overstock occurs when inventory levels significantly exceed expected demand.

Overstock Risk ≈ DOH > Threshold

This helps identify products that may lead to excess holding costs or spoilage.

Holding Cost Proxy

Inventory holding costs are estimated using total inventory value.

Holding Cost Proxy = Inventory Value * Holding Rate

Higher inventory levels lead to higher carrying costs.

Lost Sales Proxy

Lost sales occur when customer demand cannot be fulfilled due to stockouts.

Lost Sales = True Demand − Units Sold

Reducing lost sales directly improves revenue and customer satisfaction.

Scope definition

Forecasting Horizon

Demand forecasts are generated for the next 28 days.

This time horizon allows the business to plan replenishment decisions for the upcoming month.

Granularity

All analysis and replenishment decisions are performed at the store–SKU level.
This level of granularity ensures that:

- each store's demand patterns are considered
- replenishment decisions are tailored to specific products and locations.

Stakeholder questions

1. Which store–SKU combinations are at the highest risk of stockouts?

2. How accurate are the demand forecasts across different product categories?

3. What is the expected improvement in service level and lost sales reduction after applying the new replenishment policy?

4. Which stores contribute the most to overall stockout occurrences?

5. Are there products that are at risk of overstock or excess inventory?

6. How should safety stock and reorder points be set to balance availability and holding costs?

7. What is the projected business impact of implementing the optimized replenishment strategy over the next 30 days?